BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6485

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The geometry of SO_2 molecule is:

(A) Angular
(B) Linear
(C) Tetrahedral
(D) Trigonal pyramid

Q2. Which one pair is isomorphic in nature:

(A) NaCl, KBr
(B) $CaCl_2$, $CaCO_3$
(C) NaF, $MgCl_2$
(D) Na_2CO_3, $MgCO_3$

Q3. When water freezes at 0°C, its density decreases due to:

(A) Cubic structure of ice
(B) Empty spaces present in structure of ice
(C) Change of bond length
(D) Change of bond angle

Q4. Equal masses of methane and oxygen are mixed in an empty container at 25°C. The fraction of total pressure exerted by oxygen is:

(A) 1/3
(B) 8/9
(C) 1/9
(D) 16/17

Q5. Pressure remaining constant at which temperature the volume of a gas will become twice of what it is at 0°C?

(A) 546°C
(B) 546K
(C) 200°C
(D) 273K

Q6. Solvent extraction is an equilibrium process and it is controlled by:

(A) Law of mass action
(B) Distribution law
(C) The amount of solvent used
(D) The amount of solute

Q7. A beaker contains 9g of water. The number of H-atoms in it is:

(A) N^A of atoms
(B) $2 \times N^A$ of atoms
(C) $1/2\ N^A$ of atoms
(D) $3 \times N^A$ of atoms

Q8. 27g of Al will react completely with how much mass of O_2 to produce Al_2O_3?

(A) 8g of oxygen
(B) 20g of oxygen
(C) 24g of oxygen
(D) 32g of oxygen

Q9. The rate of reaction:

(A) Increases as the reaction proceeds
(B) Decreases as the reaction proceeds
(C) Remains constant as the reaction proceeds
(D) May decrease or increase as the reaction proceeds

Q10. The properties of substances which depend solely on number of particles present is known as:

(A) Additive properties
(B) Constitutive properties
(C) Additive and constitutive properties
(D) Colligative properties

Q11. Stronger the reducing agent, greater is the:

(A) Oxidation potential
(B) Reduction potential
(C) Redox potential
(D) Electromotive force of cell

Q12. A solution with pH = 2 is more acidic than a solution with pH = 6 by a factor of:

(A) 4
(B) 8
(C) 1000
(D) 10000

Q13. The value of Δn for the given equilibrium $N_2 + 3H_2 \rightleftharpoons 2NH_3$ is:

(A) –2
(B) +2
(C) +1
(D) +4

Q14. For a given process, the heat changes at constant pressure (q_p) and at constant volume (q_v) are related to each other as:

(A) $q_p = q_v$
(B) $q_p > q_v$
(C) $q_p < q_v$
(D) $q_p = q_v/2$

Q15. In ground state of an atom, the electron is present:

(A) In the nucleus

(B) In the second shell

(C) Nearest to the nucleus

(D) Farthest from the nucleus

Q16. Which is correct?

(A) Idea of presence of neutron in an atom was provided by Chadwick

(B) Neutron was discovered by using radioactive Beryllium

(C) Fast neutrons having energy 1.2ev

(D) Slow neutrons have energy above 1ev

Q17. The planer structure of BF_3 can be explained by the fact that BF_3 is:

(A) sp-hybridized

(B) sp^2-hybridized

(C) sp^3-hybridized

(D) dsp^2-hybridized

Chemistry (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 6485

Note: SECTION - I

Q2. (Marks 16) Write short answers of any EIGHT parts.

(i) Calculate mass in kilograms of 2.6×10^{20} molecules of SO_2.

(ii) Calculate mass in grams of 5.136 moles of Ag_2CO_3

(iii) Calculate mass in grams of 2.74 moles of $KMnO_4$

(iv) Define sublimation. Name two compounds which can be sublimed.

(v) Define (a) Solvent extraction (b) R^g value.

(vi) Derive the units for gas constant 'R' in general gas equation when pressure is in atmosphere and volume in dm^3.

(vii) Briefly discuss general gas equation.

(viii) Describe centigrade scale of thermometry.

(ix) Write two applications of Dalton's law of partial pressure.

(x) Define fractional distribution. Give one example.

(xi) What is non ideal solution? Give one example.

(xii) Define colligative properties. Name four colligative properties.

Q3. (Marks 16) Write short answers of any EIGHT parts.

(i) Write six crystallographic elements of a tetragonal crystal system.

(ii) Explain crystal lattice briefly.

(iii) Define transition temperature giving one example.

(iv) Explain cleavage planes.

(v) Cathode rays are material particles. Explain it.

(vi) Write any two properties of neutron.

(vii) Explain continuous spectrum briefly.

(viii) Define atomic absorption spectrum giving one example.

(ix) Give one difference between reversible and irreversible reactions.

(x) Define pH and pOH.

(xi) Define instantaneous rate and average rate of a reaction.

(xii) Define order of a reaction giving one example.

Q4. (Marks 12) Write short answers of any SIX parts.

(i) What is basic assumption of VSEPR theory.

(ii) Define coordinate covalent bond. Give example.

(iii) Define electron affinity. Give example.

(iv) Why NH_3 is a pyramidal molecule?

(v) What is system and surrounding?

(vi) Define enthalpy of combustion. Give example.

(vii) What is electrochemistry?

(viii) What is electrolytic conduction?

(ix) How electrochemical series is used to calculate voltage of cell? Give example.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 04) When lime stone is roasted, quicklime is produced according to following equation: $CaCO_3(g) \rightarrow CaO(g) + CO_2(g)$. The actual yield of CaO is 2.5kg when 4.5kg of lime stone is roasted. Find its percentage yield.

Q5(b). (Marks 04) Define and explain factors affecting the London forces.

Q6(a). (Marks 04) How Dalton's law of partial pressure calculates the partial pressure of a gas?

Q6(b). (Marks 04) Explain measurement of e/m value of electron.

Q7(a). (Marks 04) Define covalent bond. Write its types with reference to polar covalent bond.

Q7(b). (Marks 04) What is the first law of thermodynamics? How does it explain that $q_v = \Delta E$?

Q8(a). (Marks 04) When 1.00 mole of steam and 1.00 mole of carbon monoxide are allowed to reach equilibrium, 33.3% of the equilibrium mixture is hydrogen. Calculate the value of K_p. State the units of K_p.

Q8(b). (Marks 04) Explain how Arrhenius equation tells us the effect of temperature on the rate constant.

Q9(a). (Marks 04) Explain the measurement of freezing point by Beckmann's freezing point apparatus.

Q9(b). (Marks 04) Define electrochemical series. Write its two applications.

GROUP – II

Chemistry (Objective) Group - II

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6486

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Bohr model of atom is contradicted by:

(A) Plank's quantum theory
(B) Dual nature of matter
(C) Heisenberg's uncertainty principle
(D) All of these

Q2. Quantum number values for 2p orbitals are:

(A) $n = 2, \ell = 1$
(B) $n = 1, \ell = 2$
(C) $n = 1, \ell = 0$
(D) $n = 2, \ell = 0$

Q3. Which is a pseudo solid?

(A) CaF_2
(B) Glass
(C) NaCl
(D) All of these

Q4. Which is not an isomorphic pair?

(A) $NaNO_3$, KNO_3
(B) MgO and NaF
(C) K_2SO_4 and $K_2Cr_2O_7$
(D) NaF and $CaCl_2$

Q5. Number of molecules in one dm^3 of water is close to:

(A) $6.02/22.4 \times 10^{23}$
(B) $12.04/22.4 \times 10^{23}$
(C) $18/22.4 \times 10^{23}$
(D) $55.6 \times 6.02 \times 10^{23}$

Q6. Equal masses of methane and oxygen are mixed in an empty container at 25°C. The fraction of total pressure exerted by oxygen is:

(A) 1/3
(B) 8/9
(C) 1/9
(D) 16/17

Q7. The comparative rates at which the solutes move in paper chromatography depend on:

(A) The size of paper
(B) R^g values of solutes
(C) Temperature of the experiment
(D) Size of the chromatographic tank used

Q8. The ratio of actual yield to theoretical yield multiplied by 100 is called:

(A) Complex yield
(B) Experimental yield
(C) %age yield
(D) None of these

Q9. The calculation based on balanced chemical equation is called:

(A) Complex calculation
(B) Stoichiometric calculation
(C) Non-stoichiometric calculation
(D) None of these

Q10. The unit of the rate constant is the same as that of the rate of reaction in:

(A) First order reaction
(B) Second order reaction
(C) Zero order reaction
(D) Third order reaction

Q11. If the salt bridge is not used between two half cells, then voltage:

(A) Decrease rapidly
(B) Decrease slowly
(C) Does not change
(D) Drops to zero

Q12. The pH of buffers can be calculated by:

(A) Henderson equation
(B) Nerst equation
(C) Kinetic equation
(D) Arrhenius equation

Q13. Less soluble $KClO_3$ is precipitated from its solution by common ion effect on adding:

(A) HCl
(B) KCl
(C) H_2S
(D) NaCl

Q14. For which system does the equilibrium constant, Kc has units of (concentration)$^{-1}$:

(A) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (B) $H_2 + I_2 \rightleftharpoons 2HI$

(C) $2NO_2 \rightleftharpoons N_2O_4$ (D) $2HF \rightleftharpoons H_2 + F_2$

Q15. If an endothermic reaction is allowed to take place very rapidly in the air, the temperature of the surrounding air:

(A) Remains constant (B) Increases

(C) Decreases (D) Remains unchanged

Q16. Octet rule is not followed in the formation of:

(A) NF_3 (B) CF_4

(C) CCl_4 (D) PCl_5

Q17. Which species has unpaired electrons in antibonding molecular orbitals?

(A) O_2^{2+} (B) N_2^{2-}

(C) B_2 (D) F_2

Chemistry (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 6486

> Note: SECTION - I

Q2. (Marks 16) Write short answers of any EIGHT parts.

(i) No individual neon atom has a mass of 20.18 amu.

(ii) What is a limiting reactant?

(iii) What is percentage yield? Give its importance.

(iv) Define sublimation. Give example.

(v) What is solvent extraction?

(vi) Derive Avogadro's Law from KMT.

(vii) Give characteristics of plasma.

(viii) What is law of distribution of velocities?

(ix) What is centigrade scale of temperature?

(x) Give two differences between ideal and non-ideal solutions.

(xi) Define solubility and solubility curves.

(xii) Define enthalpy or heat of solution.

Q3. (Marks 16) Write short answers of any EIGHT parts.

(i) Define polymorphism by giving one example.

(ii) Define unit cell. Write dimensions of unit cell.

(iii) What is difference between crystal and crystallite?

(iv) Why H_2O is liquid and H_2S is gas at room temperature?

(v) Why the positive rays are also called canal rays?

(vi) Why alpha rays are bounced back in Rutherford experiment?

(vii) Calculate mass of electron by using e/m value.

(viii) What is difference between orbit and orbital?

(ix) Define acidic and basic buffers.

(x) Define common ion effect by giving one example.

(xi) Define rate determining step by giving one example.

(xii) Define energy of activation.

Q4. (Marks 12) Write short answers of any SIX parts.

(i) How electronegativity helps us to understand the nature of bond?

(ii) Differentiate between ionic and covalent bond.

(iii) Why some covalent bonds are polar while others are non-polar?

(iv) What is oxonium ion? How it is formed?

(v) What is enthalpy?

(vi) What is standard enthalpy of a reaction?

(vii) Calculate oxidation number of manganese in $KMnO_4$.

(viii) Calculate oxidation number of sulphur in sulphate ions.

(ix) Define electrode potential.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 04) Calculate the number of grams of K_2SO_4 and water produced when 14g of KOH are reacted with excess of H_2SO_4. $2KOH + H_2SO_4 \rightarrow K_2SO_4 + 2H_2O$

Q5(b). (Marks 04) Explain the following properties of crystalline solids. Give one example in each case: (i) Anisotropy (ii) Symmetry (iii) Polymorphism (iv) Habit of a crystal

Q6(a). (Marks 04) Derive general gas equation for one mole of a gas from gas laws at S.T.P.

Q6(b). (Marks 04) What is spectrum? Explain atomic emission spectrum and atomic absorption spectrum.

Q7(a). (Marks 04) Give the postulates of VSEPR theory.

Q7(b). (Marks 04) Explain these terms: (i) Standard heat of neutralization (ii) Standard enthalpy of solution

Q8(a). (Marks 04) $N_2(g)$ and $H_2(g)$ combine to give $NH_3(g)$. The value of Kc in this reaction at 500°C is 6.0×10^{-2}. Calculate the value of K_p for this reaction.

Q8(b). (Marks 04) (i) Define activation energy and activated complex. (ii) What is meant by specific rate constant?

Q9(a). (Marks 04) Explain Landsberger's method for the measurement of boiling point elevation.

Q9(b). (Marks 04) Define electrochemical series. Write its any two applications.